



Herbal sun protection agents: Human studies

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Abstract Topical sunscreens are the mainstay for protection from ultraviolet (UV) radiation. With skin cancer rates on the rise and great interest in reversing or preventing the effects of photoaging, new molecules with potential to defend against UV damage have received a great deal of attention. Specifically, there is a growing interest in herbal substances that offer protection against the damaging effects of UV rays. Herbal substances may work as adsorbents of the UV rays and antioxidants and potentially have few side effects. Many of them have shown the potential to protect from UV rays in *in vitro* studies and animal models; however, only a limited number of human studies were conducted which we discuss in the current review. Among the most studied herbal substances that have proven photoprotective activity are green tea extract, carotenoids, and *Polypodium leucotomos* extract (PLE). They have been shown to increase minimal erythema dose and improve signs of photodamage. PLE has been shown to be helpful in holistic treatment of several conditions, including polymorphous light eruption, solar urticaria, and melasma; it also may be used as an adjuvant to the UVB treatment of vitiligo and photodynamic therapy of actinic keratosis.

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Introduction

Sunscreens protect against sunburn and skin cancer by reflecting, absorbing, or scattering ultraviolet (UV) rays. The first use of sun protecting creams containing salicylates was recorded in the 1920s, whereas petrolatum was used as an active ingredient through the 1930s and 1940s.¹ Para-amino-benzoic acid (PABA) sun protective activity was discovered in 1940s, but it is rarely used nowadays due to a greater risk of allergic contact dermatitis.²

Minerals, such as titanium dioxide, became available in the 1950s.³ These early sunscreens were the first to offer protection against UVB. As information became available about the role of UVA in sun damage, newer agents were employed

for sun protection.⁴ Sunscreens generally may be placed in one of two categories:

1. Physical sunscreens, containing zinc oxide and titanium dioxide
2. Chemical sunscreens, containing organic PABA, cinnamates, salicylates, and benzophenones.⁵

There is a growing interest in herbal substances that offer protection against the damaging effects of UV rays. They potentially may cause fewer side effects compared with the chemical sunscreens and are attractive for patients who desire to use natural products. Photoprotective herbal substances have been studied both as standalone products as well as ingredients in topical formulations and oral supplements. Unfortunately, great variation exists in the methodology of the studies, and there are no unifying guidelines on the relative sunprotective potential of various herbs. Although numerous

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studies exist evaluating benefits on herbal extracts *in vitro*, there are limited data evaluating their efficacy in human studies. We present an overview of the topical and oral herbal sunprotective substances that potentially may be recommended to decrease the harmful effects of UVA/UVB radiation based on clinical studies.

Herbal substances as absorbents of UV

There are several mechanisms through which a plant may act as a photoprotective agent. Absorption of the UV energy by the photon and subsequent emission of less harmful energy is one of the mechanisms. Herbal extracts present a mixture of molecules that are able to absorb visible light as well as UV light of different lengths.⁶ Polyphenolic substances, flavones, flavonoids, tannins, and lipid fractions from plants have been shown to be able to absorb visible light and UVB in multiple studies.⁷ These include plant extracts that may act as natural UV filters: Antarctic plants (*Deschampsia antarctica*, *Colobanthus quitensis*, *Polytrichum juniperinum*), *Buddleja cordata*, *Coffea* genus, *Commiphora mukul*, *Crataegus pentagyna*, *Feijoa sellowiana*, *Schinus terebinthifolius*, and some lichen species.^{8–14}

One of the few *in vitro* studies of sun filtering capacity of natural substances evaluated boldine (alkaloid from the *Peumus boldus* Molina tree) (Figure 1) and aromatic compounds from Chilean lichens (usnic acid, 1-chloropannarine, epiphorelic acid I and II, and calicin) versus Nivea Sun Spray® sunscreen.¹³ Prepared solutions of natural substances and Nivea Sun Spray® with sun protection factor (SPF) of 5 were applied to the back of five volunteers and irradiated with UV. UV protection factor was measured using reflector and compared with the nonirradiated skin. Usnic acid had a highest SPF (average SPF 4.1, with the range of 3.6 to 5), followed by boldine (average SPF 3.4, with the range of 2.3 to 4.4). These results were comparable to the average measured SPF for Nivea Sun Spray®, which was 4.2, with the range of 3.6 to 5.

Herbal substances as antioxidants

UV light is known to produce reactive oxygen species. The antioxidant activity of herbal substances is another very important mechanism for sunlight protection. Phenols, flavonoids, phytoalexins, phenolic compounds, cyanidin 3-glycoside, anthocyanins, hydrolizable tannins, soy isoflavons, GAMMA-oryzanol, inositol, and fatty acids are potent natural antioxidants.⁶ Quercetin and rutin, members of the flavonoid family, have been shown *in vitro* to be as effective as sunscreens containing homosalate, a common ingredient of sunscreen.¹⁵ Silymarin, an extract of the milk thistle plant consisting of a mixture of three flavonoids, has been shown to have significant antitumor effects and a reduction in oxidative stress in murine studies as well as reduced UV-induced cell damage and apoptosis in mice.^{16,17}

Boldo

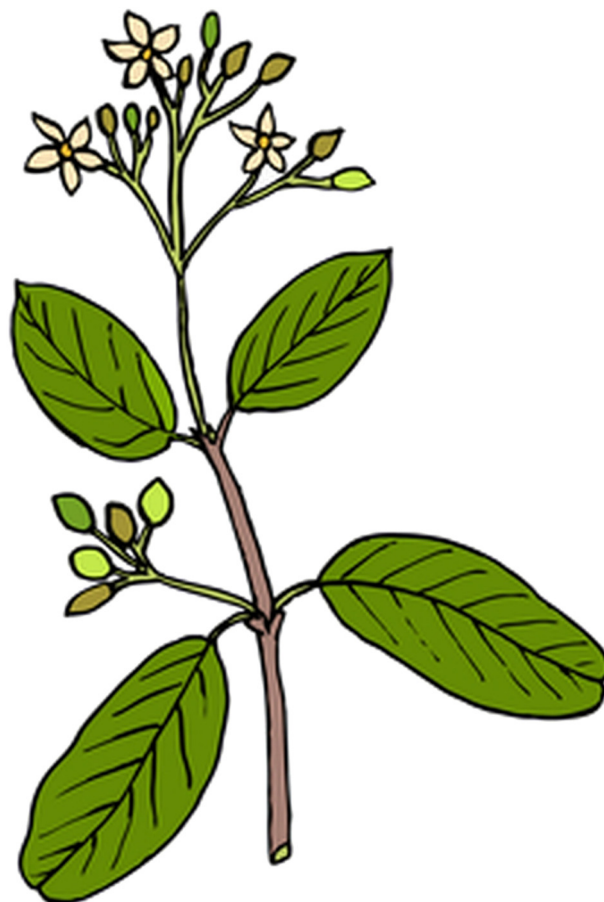


Fig. 1 Boldo (*Peumus boldus* or Molina tree)—a tree native to the central area of Chile. Retrieved from www.dreamstime.com; courtesy Foxyliam.

Turmeric

A number of studies have investigated the role of turmeric's in skin protection. Curcumin, isolated from the rhizome of turmeric (Figure 1), is commonly used as both a spice and coloring agent. Curcumin has been shown to decrease UV-induced apoptotic changes in *in vitro* studies using human keratinocytes and human epidermoid carcinoma cells.^{18,19} Murine *in vivo* studies have shown the ability of curcumin to decrease the tumor growth of squamous cell carcinoma. This has been supported by preclinical human studies.^{20–22} Although research thus far has been intriguing, further studies are required to determine how to best use curcumin's properties to elicit adequate sun protection.

Pomegranate extract

Pomegranate extract is obtained from various parts of the fruit *Punica granatum*, including the juice, seed, and peel. In particular, the phenolic components have potent antioxidant activity.²³ Topical application of the peel extract has been shown to restore catalase, peroxidase, and superoxide dismutase enzyme activities *in vivo*.²⁴ *In vitro* studies have shown pomegranate extract to protect against UVA- and UVB-induced damage.^{25,26} Murine studies have shown a protective role of pomegranate in skin cancer development.²⁷ Although pomegranate is widely available in topical formulations for its antioxidant properties, human studies are limited.²⁸

Resveratrol

Resveratrol is another agent that has been studied extensively. It is a phytoalexin found in grapes, red wine, and other fruits that has been shown to have antitumor effects in a number of murine models, as well as against UVB-induced skin carcinogenesis in mice.²⁹ The excitement of these early preclinical studies has spawned great interest in the molecule, but no significant clinical data and therefore no recommendations exist to increase the daily intake of resveratrol beyond what is available through the average diet.³⁰

Other herbs evaluated in *in vitro* and *in vivo* preclinical studies and those that have proven to have antioxidant activity include *Camellia sinensis*, *Castanea sativa*, citruses, strawberries, *Ginkgo biloba*, *Glycine max*, and *Oryza sativa*.^{31–35}

Green tea extract

Camellia sinensis leaves and leaf buds are used to produce green tea (Figure 2). Green tea extract (GTE) is consumed in many regions of the world for its anti-inflammatory effects. Epigallocatechin-3-gallate (EGCG) is the main phenol component that is able to block oxidative stress, expression of metalloproteinases, induces tumor carcinogenesis and angiogenesis.^{31,36,37} Topical application of EGCG may decrease UV-induced infiltration of the skin by lymphocytes, hydrogen peroxide, and nitric oxide in human subjects.^{38–40} (See Figs. 3 and 4).

In a double-blind placebo-controlled study, 56 participants were randomized to receive 250 mg GTE capsule and placebo. After 6 months of clinical evaluation by a dermatologist for photodamage, the GTE group showed a reduction of damage ($P = 0.02$). Erythema and telangiectasias were significantly decreased in the GTE group at 6 and 12 months from baseline. These differences were not sustained at the 24-month time point, but the researchers suggested that photoprotective effect starts earlier in the GTE group. Using Visia Complexion Analysis, lessening of brown spots and UV spots was registered at 1 year in GTE group, but no change of other parameters was seen at any timeline (texture unevenness, wrinkles, pores). Also, no histopathologic differences between GTE and placebo groups (overall solar damage, solar elastosis,

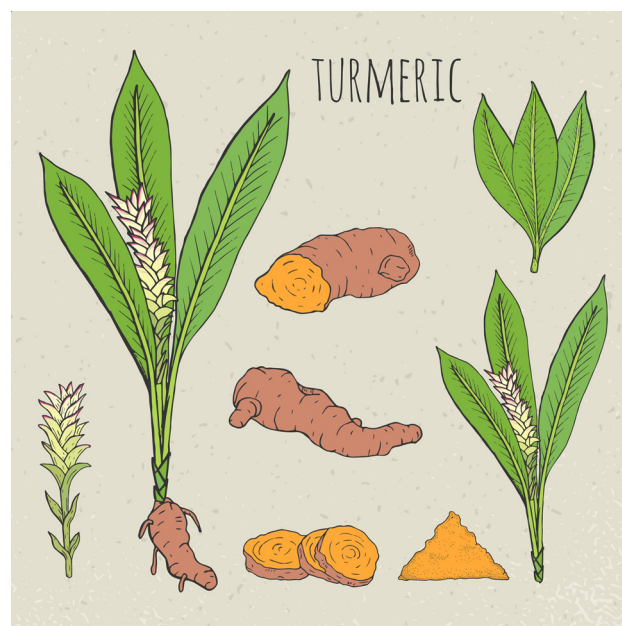


Fig. 2 Turmeric. Retrieved from www.dreamstime.com; courtesy Ernest Akayeu.

dermal mucin, epidermal melanin, perivascular inflammation, epidermal thickness, dermal collagen, or dermal elastin) were observed.⁴¹

In another double-blind study, 30 women drank 1 L of green tea daily and 30 other women consumed a neutral beverage; improvement of different skin parameters was measured by 12th week. The following have been documented: decrease of resistance of the inner forearm skin to move toward an applied vacuum by 21%; increase of biologic elasticity by 3.9%; increase of skin density by 7.7%; improvement of roughness, volume, and decline of scaling; increase of skin hydration up to 17%; decrease of transepidermal water loss by 12%; and increase of cutaneous blood flow by 29%.⁴²

Other studies have found no beneficial effects for GTE, absence of difference in minimal erythema dose (MED) and visible erythema in GTE, and no difference in the numbers of CD3+ T lymphocytes in GTE and control groups.⁴³

Carotenes are natural antioxidants that are present in some fruits and vegetables and were noted to have photoprotective activity long time ago. In early clinical studies, supplementation with carotene was shown to be effective in preventing photodermatoses; helping in treating some photo-related conditions, including porphyria; and decreasing MED.^{44–47} Tomatoes and tomato paste are the excellent source of carotenes. Consumption of 40 g tomato paste per day with 10 g olive oil was helpful to decrease erythema values as well as increase MED in healthy subjects.^{48,49} After 3 months of supplementation, there was a significant increase in procollagen I as well as a decrease in metalloproteinase-1 (MMP-1), after UVR in the group receiving tomato paste.⁴⁹



Fig. 3 *Camellia sinensis*, used to produce green tea. Retrieved from www.dreamstime.com; courtesy Luisa Vallon Fumi.

***Polypodium leucotomos* extract**

Some plants have synergistic photoprotective effects and multiple mechanisms of photoprotection. *Polypodium leucotomos* extract (PLE), which is extracted from a fern growing in tropical regions of the Americas, is the most well-studied oral supplement with photoprotective qualities. Many *in vitro* studies have shown the ability of PLE to shield human keratinocytes from deleterious effects of UV exposure. PLE treatment of keratinocytes has been reported to decrease the production of tumor necrosis factor- α and nitric oxide, and decrease the activation of nuclear factor- κ B (NF κ B), all of which have been implicated in UVB-induced skin carcinogenesis.^{50,51} More recently, PLE has been shown to protect against UVB-induced damage in an *in vitro* model of human epidermis with reduction in sunburnt cells and formation of cyclobutane pyrimidine dimers (CPD).⁵² Preclinical studies of mice have also had promising results. Daily oral administration of PLE delayed tumor development and caused an increase in p53 expression in *hairless mice*. The increased p53 expression is thought to be protective by two possible mechanisms, including increasing apoptosis or repairing damaged DNA.⁵³



Fig. 4 *Polypodium leucotomos*, a fern with photoprotective activity. Retrieved from www.dreamstime.com; courtesy Luisa Vallon Fumi.

The results from the *in vitro* and *in vivo* studies have generated a great deal of interest and have led to a number of clinical studies. Several commercially available brands of PLE are available in the United States and Europe (Heliocare® and Fernblok®). PLE was shown to be effective in healthy subjects in reducing erythema and increasing MED, patients with solar urticarial, polymorphous light eruption (PMLE), melasma, vitiligo, melanoma, and actinic keratosis (AKs).

Administration of two doses of PLE orally before UV exposure (7.5 mg/kg body weight) causes significant reduction

of visible erythema and an increase of MED in pretreated healthy subjects.^{54,55} Additionally, biopsies show decreased UV-related epidermal damage (maturation disarray, microvesiculation, and vacuolization of keratinocytes) in the group pretreated with PLE as well as reduced expression of CPD, Ki-67, cyclin D1, proliferating cell nuclear antigen (PCNA), and cyclooxygenase-2 (COX-2).^{54,55} Although not statistically significant, pretreatment with PLE in healthy volunteers showed a strong trend to decrease UVA-induced photoaging marker. This marker is a 4977-base-pair-long mitochondrial DNA and is a common deletion induced by and thought to be a marker of chronic UVA radiation in fibroblasts and keratinocytes.⁵⁶

Photoprotective activity of PLE was assessed in a study of 57 individuals with PMLE and solar urticaria. When study participants consumed 480 mg per day of PLE orally and exposed themselves to sunlight, they had significant reduction of skin reaction and subjective clinical manifestations.⁵⁷ In another study in patients with PMLE, PLE completely blocked PMLE lesion development in 9 patients out of 30 after UVA photoprovocation (30%) and in 5 of the 18 subjects after UVB photoprovocation (28%). Of those patients who were not completely protected with PLE, partial protection occurred. The mean number of exposures required to produce PLE lesions increased from 1.95 to 2.62 for UVA ($P = 0.05$) and from 2.38 to 2.92 for UVB ($P = 0.047$).⁵⁸

PLE has been also used in combination with phototherapy for vitiligo and has been shown beneficial in managing melasma, where it was used as an adjuvant for sunscreen. One study evaluated PLE in patients with melasma in combination with topical sunscreen. Forty subjects were randomized to receive either 240 mg doses of oral PLE or placebo 3 times daily for 12 weeks. Both groups improved at weeks 6 and 12, but no significant difference was noticed between both control and placebo group that only was using sunscreen.⁵⁹ A randomized double-blind study of patients with vitiligo undergoing NB-UVB therapy showed that repigmentation in the head and neck area was higher in the group taking PLE 250 mg 3 times per day compared with placebo group.⁵⁴

The photoprotective ability of PLE to increase MED was studied in patients with sporadic and familial melanomas. Sixty-one patients with familial and/or multiple melanoma were exposed to different doses of artificial UVB radiation with 1080 mg PLE supplementation and without. MED was increased in the treatment groups, but proved to be particularly helpful in cases of familial melanoma.⁶⁰

PLE has also been used in combination with photodynamic therapy (PDT) in treatment of actinic keratosis of scalp and seems to improve long-term PDT clearance rates of AKs.⁶¹ Thirty-four patients presenting with at least two AKs on the scalp were assigned to two groups. Both groups underwent two PDT sessions 1 week apart. The first group began oral PLE supplementation with 960 mg/day for 5 months and showed reduction of AKs in 6 months compared with the control group. Other agents that have been used in combination with PLE include pomegranate. In a recent study, the

combination of PLE and pomegranate was shown to be more effective than PLE alone in reducing erythema in healthy subjects.⁶²

No significant side effects were reported in clinical studies with PLE. No target organs or treatment-related toxicologic effects were seen in Wistar rats.⁶³ A randomized, double-blind, placebo-controlled trial determined that administration of 240 mg of PLE taken twice daily for 60 days was safe in adults. A total of 40 patients randomly received the PLE and placebo. The study participants then underwent assessments, including physical examination, vital signs, and laboratory testing, including comprehensive metabolic panel and clotting studies at baseline and at the end of the study, with no changes found in these parameters. The efficacy was determined by MED testing, UV-induced erythema intensity response, and sunburn history during the 60-day study period.

The subjects in the placebo group had a greater chance of experiencing sunburn episodes. At day 28, PLE-treated subjects showed greater likelihood of an increased MED and greater likelihood of decreased UV-induced erythema intensity.⁶⁴

Conclusions and future perspectives

In spite of increasing interest in the herbal photoprotective substances, rigorous controlled studies need to be initiated. There are many factors to consider when evaluating the efficacy of the herbal substances in clinical studies. Bioavailability and safety are only few of them. Only several herbal substances, including PLE, GTE, and carotens, have enough evidence for potential recommendation of their use in humans.

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